

Jun 24 (First meeting!)

Track 3 ideas (impact not only attendees but also urban planners)

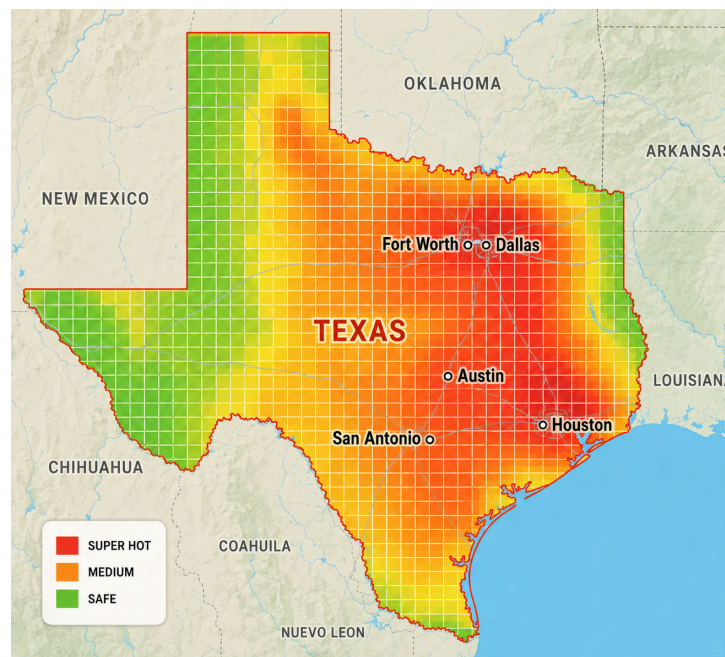
FIFA HEATSAFE AI Prototype:

- An online tool that helps both users/city professionals predict danger zones for heat, dehydration, energy usage, weather, and crowd overlap.
- **Daily Weather** dataset for climate/energy risk prediction, **Core POI Geometry** to know the location of foot traffic in a city, **Urban Heat Index** to know how much hotter a specific city is relative to nearest neighbors, daily visits to see traffic for businesses

Features/Functions:

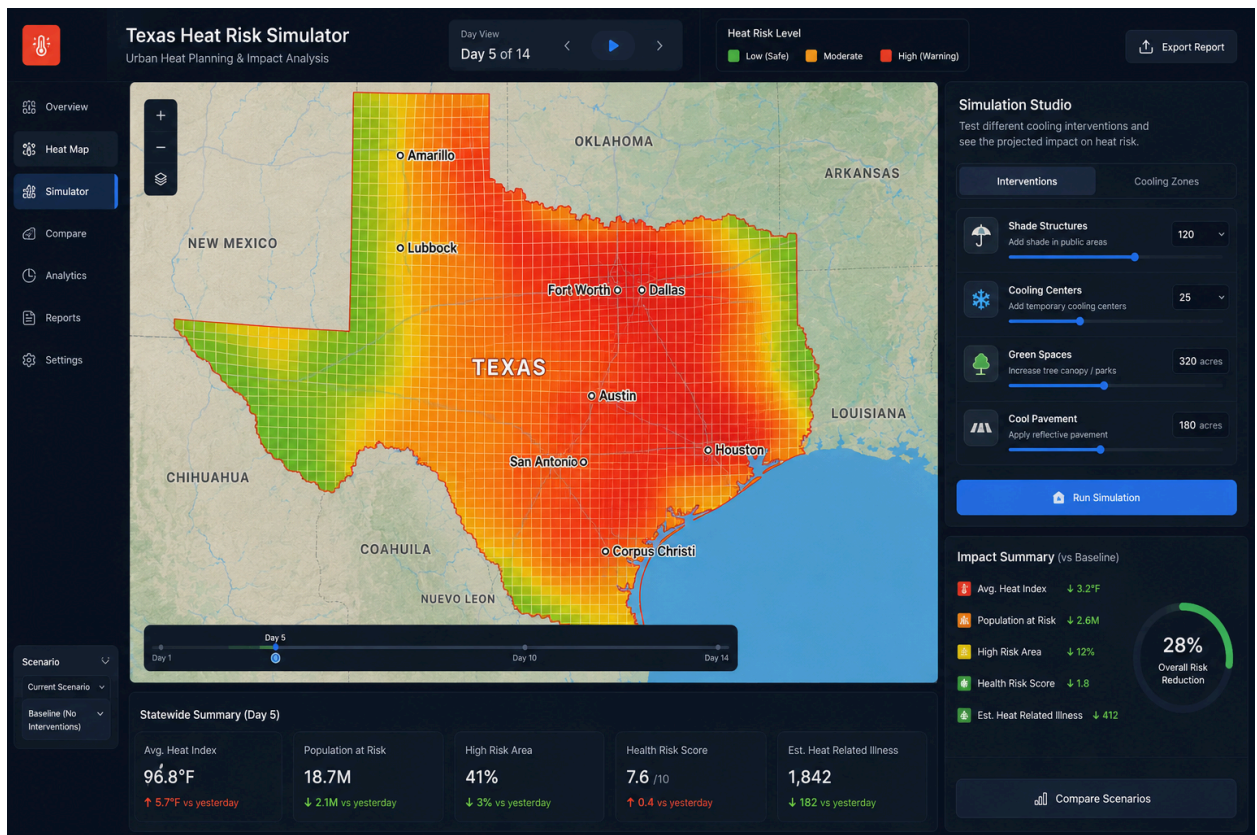
1. Maps out visitors activities, urban heat and climate risks for state based on inference from data provided on specific key cities

- An aggregate map/timeline of the chosen state (for now). With red, orange, and green cells representing the nth day avg heat, warning etc
- Heat Mapping



- Visitor Mapping
- Risk Score Mapping: Based on heat and visitor mapping
- This timeline can be extended to the future as well, maybe few days to few months forward, predicted based on past data

2. A stimulation platform that allows urban planners to try different implementations/solutions (such as adding shading structures, cooling zones) and see how those solutions improve the health risks of those visitors



TECHNOLOGY/IMPLEMENTATION

Front End

- React + TypeScript
- Some kind of stimulation engine

Back End

- Given specific weather, position, location data of a specific city; how can we infer on that data using stats/ML to predict in on other cities
- Postgres neon database (Michael will create + share access) + render backend + vercel front end
- FastAPI, Sqlalchemy, pydantic, oath2, celery, redis, JWT,
-

- **Model**

- **Statistical Technology**

- **XG Boost/Linear Regression** to learn features based on historical data and predict tomorrow's conditions (tomorrow's heat index based on humidity, speed, weather forecast, temperature, rain, etc)
 - **Random forest:**
 - **determine heat risk score formula**
 -

Database (Michael will create on 25th june, discuss more on friday)

Main Postgres Table Structure

- Basically based on the datasets they give us
- Set up the cloud database, give basic schema+sqlalchemy database creation schemes + set up the alembic datamigration tool to actively update tables + review the dataset
- Use draw.io to design our databases

- **6 Database containing columns w/ the same name as the dataset provided + the same column names, but with no data (example: “weather table”: heat, humidity, temp, etc columns**
- **Grid cell prediction table: position, metric level, color, date, etc Table w/ predicted metrics: predicted heat, predicted color, predicted etc**

Before the next meeting on Friday, June 26th

EVERYONE PLEASE TAKE A DETAILED LOOK ON THE DATASET/PPT slide / HAVE GPT SUMMARIZE IT

Specific/key datasets: Daily Weather, Core POI Geometry, Urban Heat Index

Michael:

Creating an online PostgreSQL database/setup basic app schemas.

(1-1 session w/ phat during weekend to design backend etc)

The Phat:

Map out the tech stack and decide what technologies to use for each part

Preston:

Find out the best statistical models (XG BOOST, Regression, Forest) that fit our heat prediction/inference and test it out on 1 CSV file just to show proof of inference.

Input table -> process it to only include cities in x state -> model -> inference metric (heat index metric to decide color optional: the position of this map but not necessary for now)

If have time, brief presentation on which ML model you think is best for each specific case

Zach/Gorden:

To analyze the dataset more in depth (give statistics, such as mean, sd, basic graphs of the dataset and how each column/dataset could apply. I want a simple presentation and how it applies to our topic)